

Executive Summary, Introduction, and Methodology

Final Presentation PDF • GitHub repository contains all completed notebooks and Python files.

Executive Summary: 90 model-ready launches; landing success was 66.7%; top test accuracy was 83.3%.

Introduction: predict Falcon 9 first-stage landing success because recovery can reduce launch cost.

Data Collection & Wrangling: SpaceX API and Wikipedia; cleaned records; binary Class label.

EDA/Data Visualisation & Classification: charts and SQL assessed site, orbit, payload and time; four models were evaluated.

EDA, SQL, Folium & Dash Results

EDA: landing success rose as operations matured.

SQL: four successful drone-ship recoveries were identified at 4,000–6,000 kg.

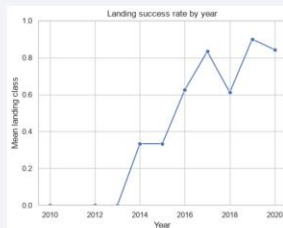
Folium: green/red outcomes and a 0.08 km proximity measurement were mapped.

Plotly Dash: CCAFS SLC 40 showed 95% successful and 5% failed launches.

Launch Success Yearly Trend

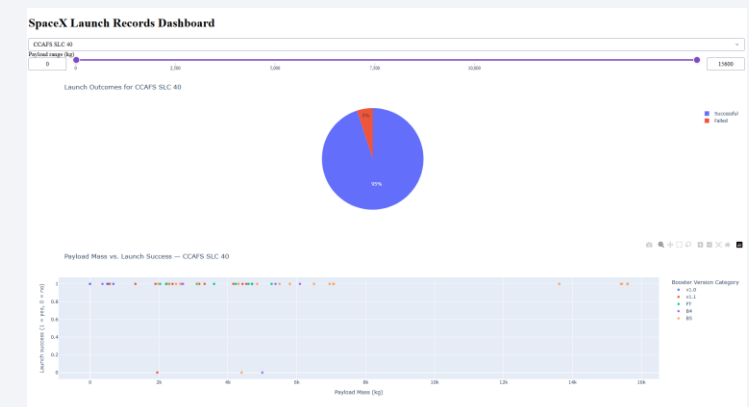
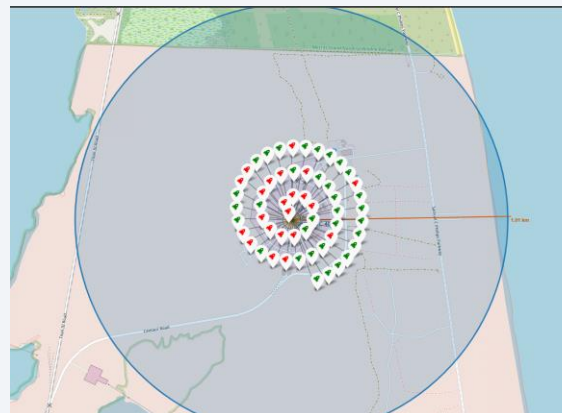
The yearly average landing-success rate trends upward as recovery operations mature.

The pattern supports including year and flight number as predictive features.



GitHub repository: <https://github.com/darizek-art/ibm-data-science-capstone>

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Prediction Results, Creativity, and Conclusion

Predictive Analysis (Classification): Logistic Regression, SVM and KNN tied at 83.3%.

Confusion matrix: 12 correct landings; 3 missed non-landings; 3 false landing predictions.

Creativity: Folium proximity and Plotly Dash linked geography, payload and outcomes.

Innovative insight: success rose with maturity; site, orbit, payload and flight number mattered.

Conclusion: useful signal, but the small test set warrants caution.

